

P61. 2

试用 MATLAB 软件编程实现追赶法求解三对角方程组的算法, 并考虑如下梯形电阻电路问题。

设电路中的各个电流 $\{i_1, i_2, \dots, i_8\}$ 满足线性方程组

$$\begin{aligned}2i_1 - 2i_2 &= V/R, \\-2i_1 + 5i_2 - 2i_3 &= 0, \\-2i_2 + 5i_3 - 2i_4 &= 0, \\-2i_3 + 5i_4 - 2i_5 &= 0, \\-2i_4 + 5i_5 - 2i_6 &= 0, \\-2i_5 + 5i_6 - 2i_7 &= 0, \\-2i_6 + 5i_7 - 2i_8 &= 0, \\-2i_7 + 5i_8 &= 0.\end{aligned}$$

设 $V = 220\text{ V}$, $R = 27\ \Omega$, 求各段电路的电流。

解：实验内容、步骤及结果

先编写追赶法函数 chap-2/thomas_algorithm.m:

```
function x = thomas_algorithm(a, b, c, d)
    % a: sub-diagonal (length n-1)
    % b: main diagonal (length n)
    % c: super-diagonal (length n-1)
    % d: right-hand side (length n)

    n = length(b);

    % Forward elimination
    for i = 2:n
        w = a(i-1) / b(i-1);
        b(i) = b(i) - w * c(i-1);
        d(i) = d(i) - w * d(i-1);
    end

    % Back substitution
    x = zeros(n, 1);
    x(n) = d(n) / b(n);
    for i = n-1:-1:1
        x(i) = (d(i) - c(i) * x(i+1)) / b(i);
    end
end
```

然后建立本题对应的三对角方程组并调用追赶法程序 chap-2/Q2.m:

```

% Define input parameters
a = -2;
a = repmat(a, 1, 7);
b = [2, 5, 5, 5, 5, 5, 5, 5];
c = a;
V = 220;
R = 27;
d = [V/R, 0, 0, 0, 0, 0, 0, 0];

% Call the function
x = thomas_algorithm(a, b, c, d);

% Display the output
disp(x);

```

实验结果分析

```

>> Q2
    8.1478
    4.0737
    2.0365
    1.0175
    0.5073
    0.2506
    0.1194
    0.0477

```

因此，各段电流近似为

$$(i_1, i_2, \dots, i_8)^T \approx (8.1478, 4.0737, 2.0365, 1.0175, 0.5073, 0.2506, 0.1194, 0.0477)^T.$$

可见电流沿梯形电阻电路逐段递减。

P61. 3

方程组的性态和矩阵条件数实验。设有线性方程组 $Ax = b$ ，其中系数矩阵 $A = (a_{ij})_{n \times n}$ 的元素分别为

$$a_{ij} = x_i^{j-1} \quad (x_i = 1 + 0.1i, \quad i, j = 1, 2, \dots, n)$$

或

$$a_{ij} = \frac{1}{i+j-1} \quad (i, j = 1, 2, \dots, n),$$

右端向量

$$b = \left(\sum_{j=1}^n a_{1j}, \sum_{j=1}^n a_{2j}, \dots, \sum_{j=1}^n a_{nj} \right)^T.$$

利用 MATLAB 中的库函数完成下列实验：

1. 取 $n = 5, 10, 20$ ，分别求出系数矩阵的 2-条件数，判别它们是否为病态阵，并说明随着 n 的增大矩阵性态如何变化。
2. 分别取 $n = 5, 10, 20$ ，解上述两个线性方程组 $Ax = b$ 。将求得的解与精确解 $x = (1, 1, \dots, 1)^T$ 作比较，并说明实验现象。
3. 取 $n = 10$ ，对系数矩阵中的 a_{22} 和 a_{nn} 增加一个扰动 10^{-8} ，求解方程组 $(A + \delta A)x = b$ ，观察解的变化。

解：实验内容、步骤及结果

为便于统一生成两类线性方程组，先编写辅助函数 chap-2/generateLinearSystem.m：

```
function [A, b] = generateLinearSystem(n, matrixType)
    % This function generates the coefficient matrix A and the vector b for
    % the linear system Ax = b. The parameter n specifies the size of the
    % matrix, and matrixType specifies the type of matrix to generate.
    %
    % matrixType can be:
    % 'polynomial' - A is generated using a_{i j} = x_i^{j-1} where x_i = 1
    %               + 0.1 * i
    % 'hilbert'    - A is generated using a_{i j} = 1 / (i + j - 1)

    if strcmp(matrixType, 'polynomial')
        % Generate polynomial matrix
        A = zeros(n);
        for i = 1:n
            x_i = 1 + 0.1 * i;
            for j = 1:n
                A(i, j) = x_i^(j-1);
            end
        end
    elseif strcmp(matrixType, 'hilbert')
        % Generate Hilbert matrix
        A = zeros(n);
        for i = 1:n
            for j = 1:n
                A(i, j) = 1 / (i + j - 1);
            end
        end
    else
        error('Invalid matrix type. Use 'polynomial' or 'hilbert'.');
    end
end
```

```
% Compute vector b
b = sum(A, 2);
end
```

(1) 计算条件数

条件数实验程序 chap-2/Q31.m:

```
% Define the values of n
n_values = [5, 10, 20];
matrixType = 'polynomial'; % or 'hilbert'

% Loop through each value of n
for n = n_values
    % Generate the linear system
    [A, b] = generateLinearSystem(n, matrixType);

    % Compute the 2-norm condition number
    cond_num = cond(A, 2);

    % Determine if the matrix is ill-conditioned
    if cond_num > 1e10 % Cond(A) >> 1
        result = 'Ill-conditioned';
    else
        result = 'Well-conditioned';
    end

    % Display the condition number and the result
    fprintf('%s, n=%d, 2-norm condition number=%%.2e, Result:%s\n',
        matrixType, n, cond_num, result);
end
```

运行结果如下:

```
>> Q31
polynomial, n = 5, 2-norm condition number = 5.36e+05, Result: Well-
conditioned
polynomial, n = 10, 2-norm condition number = 8.68e+11, Result: Ill-
conditioned
polynomial, n = 20, 2-norm condition number = 6.07e+21, Result: Ill-
conditioned
>> Q31
hilbert, n = 5, 2-norm condition number = 4.77e+05, Result: Well-
conditioned
hilbert, n = 10, 2-norm condition number = 1.60e+13, Result: Ill-
conditioned
```

```
hilbert, n = 20, 2-norm condition number = 2.11e+18, Result: Ill-  
conditioned
```

由结果可见, 随着 n 增大, 两类矩阵的 2-条件数迅速增大, 矩阵病态程度显著增强。

(2) 比较数值解与精确解

求解线性方程组并计算相对误差的程序 chap-2/Q32.m:

```
% Define the values of n  
n_values = [5, 10, 20];  
matrixTypes = {'polynomial', 'hilbert'};  
  
% Exact solution  
x_exact = @(n) ones(n, 1);  
  
% Loop through each matrix type and value of n  
for matrixType = matrixTypes  
    for n = n_values  
        % Generate the linear system  
        [A, b] = generateLinearSystem(n, matrixType{1});  
  
        % Solve the linear system  
        x_approx = A \ b;  
  
        % Compute the relative error using the infinity norm  
        rel_error = norm(x_approx - x_exact(n), inf) / norm(x_exact(n), inf  
            );  
  
        % Display the relative error  
        fprintf('%s, n=%d, Relative Error (inf-norm) = %.2e\n',  
            matrixType{1}, n, rel_error);  
    end  
end
```

运行结果如下:

```
>> Q32  
polynomial, n = 5, Relative Error (inf-norm) = 2.89e-11  
polynomial, n = 10, Relative Error (inf-norm) = 7.21e-05  
polynomial, n = 20, Relative Error (inf-norm) = 3.50e+05  
hilbert, n = 5, Relative Error (inf-norm) = 6.17e-13  
hilbert, n = 10, Relative Error (inf-norm) = 3.18e-04  
hilbert, n = 20, Relative Error (inf-norm) = 2.64e+01  
  
Warning: Matrix is close to singular or badly scaled. Results may be  
inaccurate. RCOND = 3.713214e-23.
```

```
> In Q32 (line 15): x_approx = A \ b;  
Warning: Matrix is close to singular or badly scaled. Results may be  
inaccurate. RCOND = 1.276108e-19.  
> In Q32 (line 15): x_approx = A \ b;
```

随着 n 增大, 系数矩阵的病态程度进一步提升, 造成解的相对误差明显增大; 当矩阵高度病态时, 即使理论精确解为全 1 向量, 数值计算结果也会出现严重偏离。

(3) 考察微小扰动对解的影响

扰动实验程序 chap-2/Q33.m:

```
% Define the value of n  
n = 10;  
matrixType = 'polynomial'; % or 'hilbert'  
  
% Generate the original linear system  
[A, b] = generateLinearSystem(n, matrixType);  
  
% Exact solution  
x_exact = @(n) ones(n, 1);  
  
% Solve the original system  
x_original = A \ b;  
  
% Introduce perturbation  
A_perturbed = A;  
A_perturbed(2, 2) = A_perturbed(2, 2) + 1e-8;  
A_perturbed(n, n) = A_perturbed(n, n) + 1e-8;  
  
% Solve the perturbed system  
x_perturbed = A_perturbed \ b;  
  
% Compute relative errors  
relative_error_exact = norm(x_original - x_exact(n), inf) / norm(x_exact(n),  
    , inf);  
relative_error_perturbed = norm(x_perturbed - x_exact(n), inf) / norm(  
    x_exact(n), inf);  
  
% Display the results  
fprintf('%s\n', matrixType);  
fprintf('Relative error of the original solution: %.2e\n',  
    relative_error_exact);  
fprintf('Relative error of the perturbed solution: %.2e\n',  
    relative_error_perturbed);
```

分别取 `matrixType = 'polynomial'` 和 `matrixType = 'hilbert'` 运行, 结果如下:

```
>> Q33
polynomial
Relative error of the original solution: 7.21e-05
Relative error of the perturbed solution: 3.16e-01
>> Q33
hilbert
Relative error of the original solution: 3.18e-04
Relative error of the perturbed solution: 9.77e+00
```

当 $n = 10$ 时, `polynomial` 和 `hilbert` 两类线性系统都已表现出明显病态性。此时即使只在矩阵中引入 10^{-8} 量级的微小扰动, 也会使解的相对误差显著放大, 说明病态方程组对数据误差极其敏感。